



Bearward Engineering Ltd. accepts no responsibility for any consequential damage or injury caused by the execution of any of the instructions detailed.

Bearward Engineering Ltd. Can provide technicians to advise and train personnel in line with all of our procedures, for further information please contact us at:
ben_sales@wabtec.com

Quality Sign off M RICHARDSON	Process Owner Sign off M EGGLETON	Technical Sign Off: D DOOLEY		
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1. SAFETY

- 1.1. Bearward have designed the lifting process to utilise a 4-leg lifting chain. Part of the lifting/turning procedure requires 2-legs only to be used (unused 2-legs to be hooked away for safety and to prevent potential damage). Therefore 2-legs must be able to lift the total weight of largest radiator plus transport skid, refer to instalation drawing for weight of the unit.
- 1.2. Minimum leg length of chains is 2.5m
- 1.3. Always use lifting chains at full length whenever possible and use safe working practices when setting leg angles. Never set leg angle more than 60° and work within the safe range for the chains.
- 1.4. only use suitably rated lifting equipment
- 1.5. always ensure area is clear when turning radiator
- 1.6. always turn on flat and level ground
- 1.7. do not place radiator on any support parts during turning, i.e. wooden bearers, this can subject the radiator to extra forces / strains and lead to radiator damage

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2. TORQUE SETTINGS

M6	12Nm / 9 ft-lb
M8	29Nm / 21 ft-lb
M16 Idler Bolts	210Nm / 155 ft-lb
M24	50Nm / 37 ft-lb

- 2.1. Whenever bolts are tightened, refer to these torque requirements unless otherwise stated.
- 2.2. It is recommended that any bolts or nuts removed while carrying out these instructions are replaced with new, equivalent or greater grade fixings and applying the same torques as indicated above. Failure to do so could result in premature failure of parts.
- 2.3. When replacing metric bolts they should be tightened to the above torque settings.
- 2.4. Always retrieve dropped nuts, washers and bolts. Any left in the radiator can cause irreparable damage when the system is started.

3. SHIPPING

- 3.1. For transportation purposes, including sea freight, it is recommended that the radiator is located and fixed securely to minimise the risk of movement.
- 3.2. **ATTENTION:** The fan should be disconnected from the belts and allowed to rotate freely during transportation. Failure to do so may cause irreparable damage to the bearings.



Bearward Engineering

A Wabtec Company

**LARGE RADIATOR SHIPPING HANDLING SETUP &
MAINTENANCE**

PI-10-10-00-06

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Issue/Mod: **0 5678**

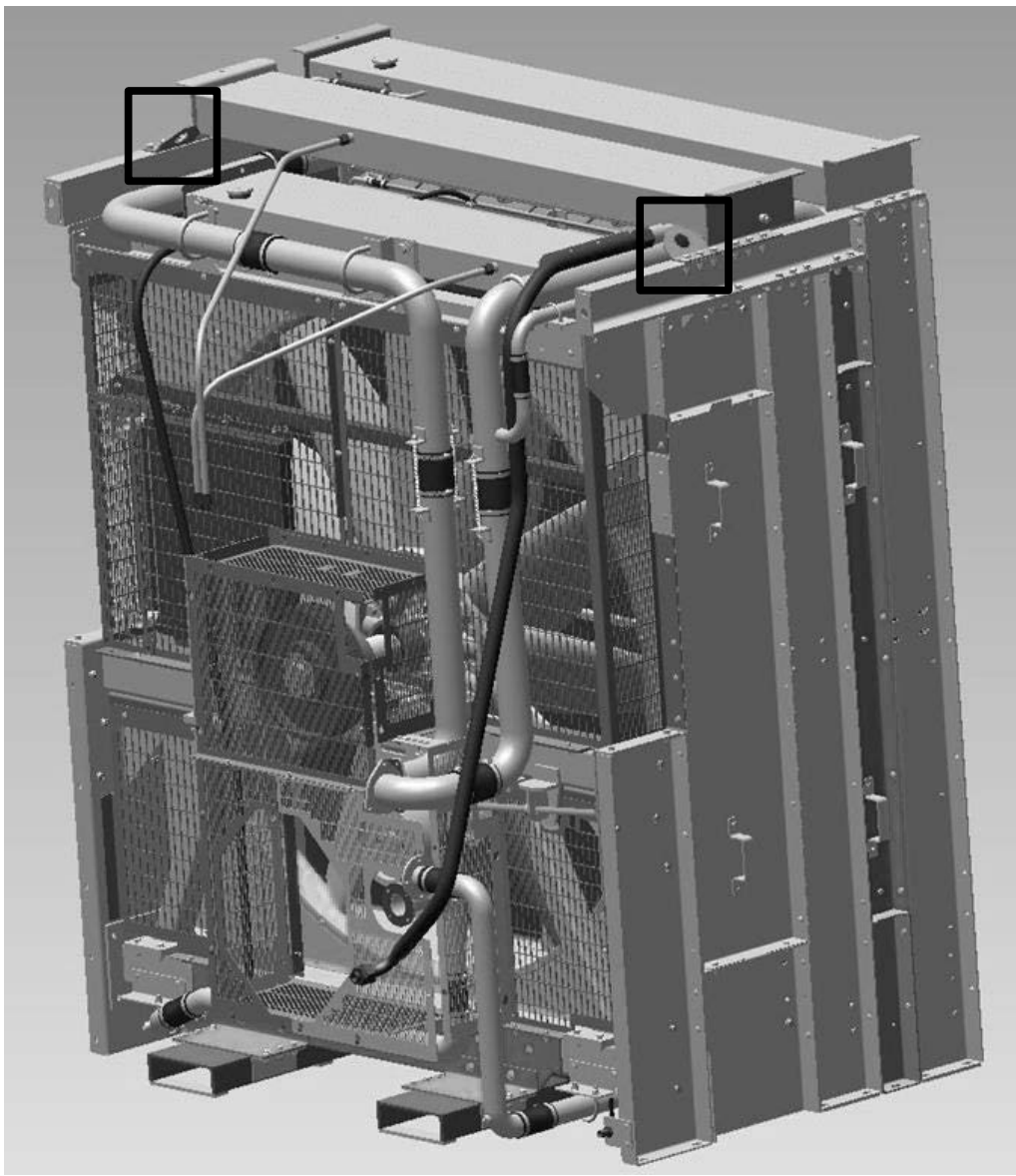
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PI-10-10-00-06-LargeRad-Ship-Maintenance

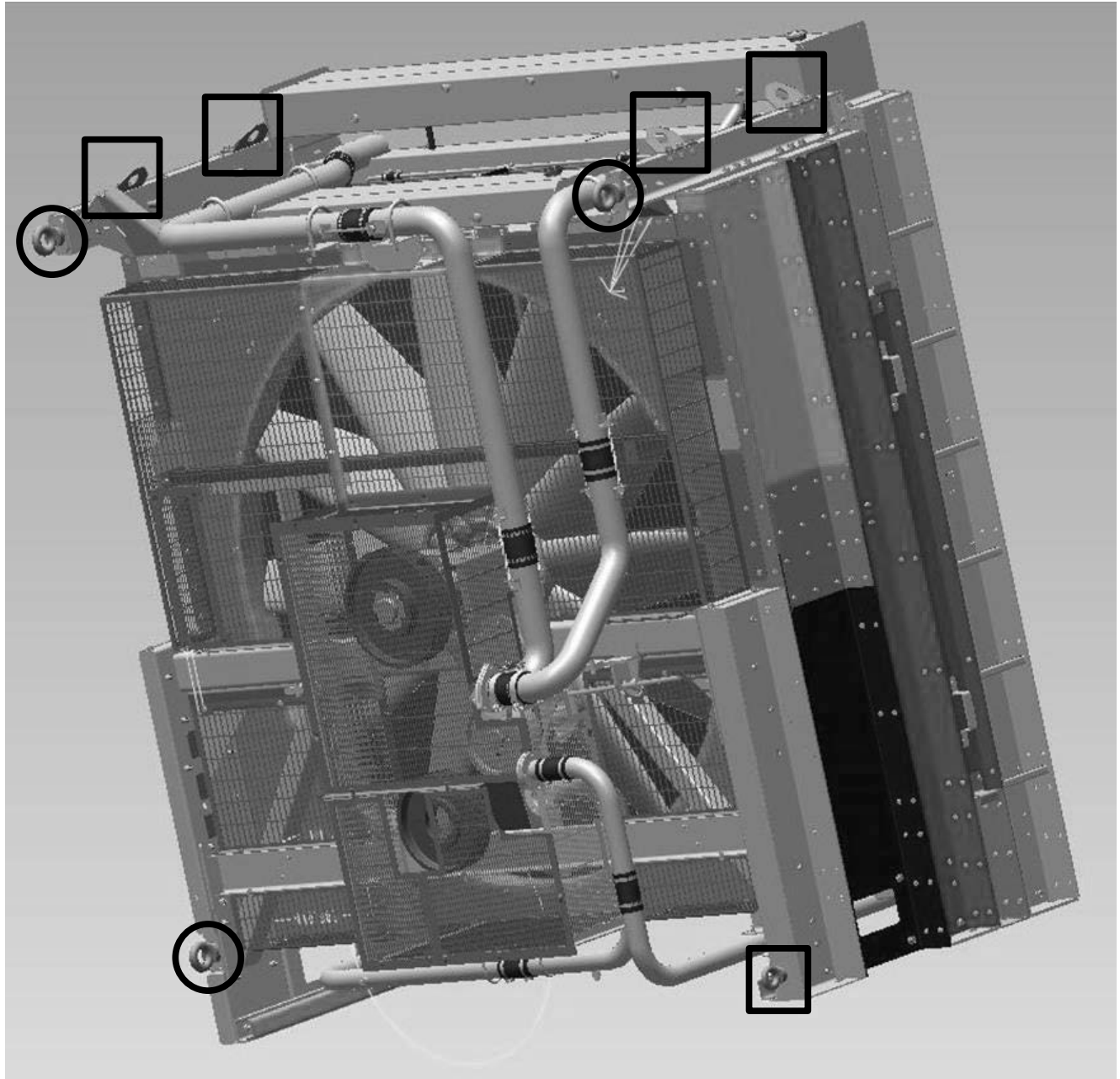
THESE ARE OUR 3 MAIN TYPES OF RADIATOR AND BELOW SHOWS THE TYPICAL LIFTING POINT AND THEIR USE

□ = Straight lifting point
○ = Turning point

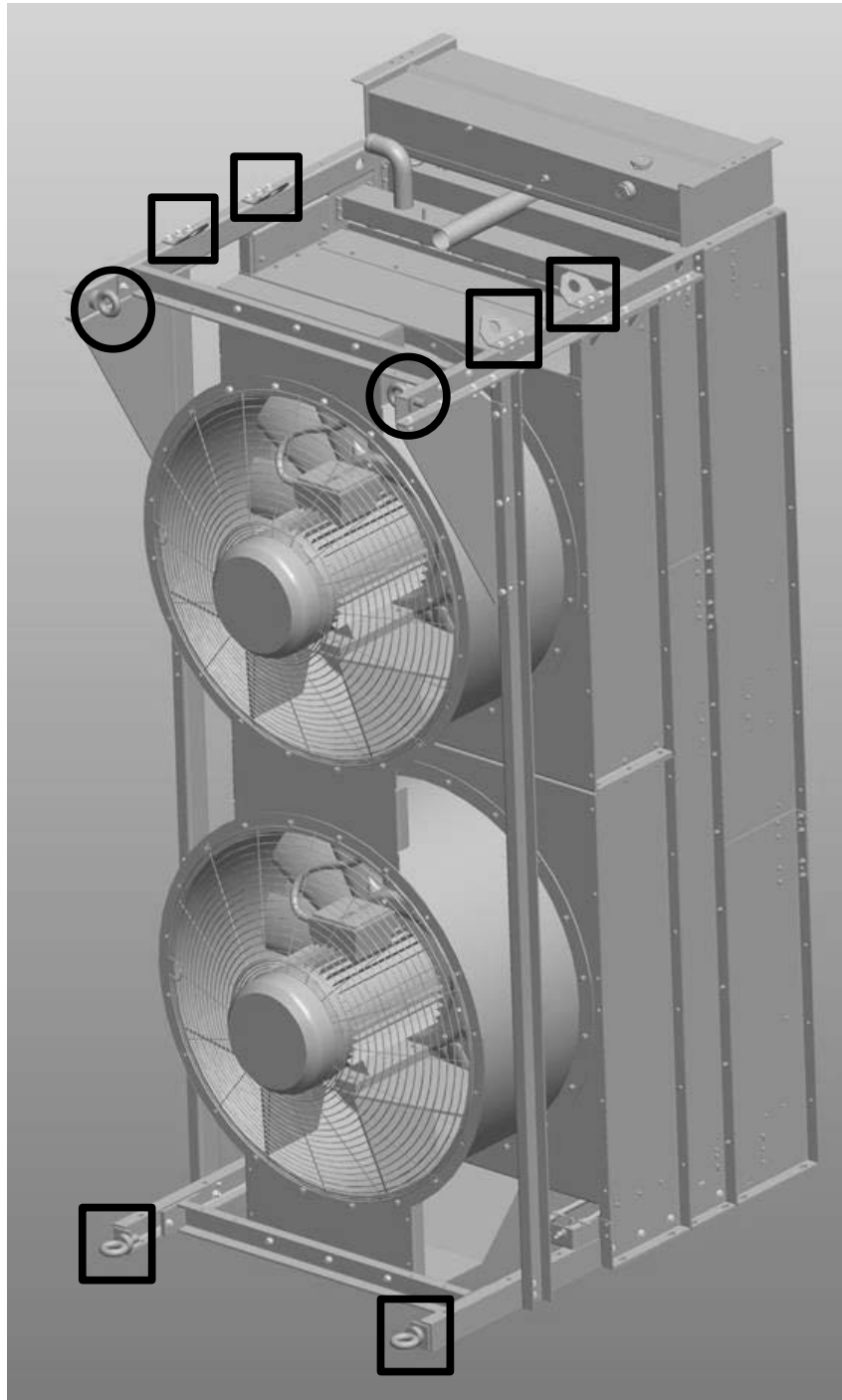
**4. TYPE 1
Shipped upright**



5. TYPE 2
Shipped on it's side



6. TYPE 3
Shipped on it's front



7. TOOLS REQUIRED

2 x 36mm Open ended spanners/wrenches.
 2 x 46mm Open ended spanners/wrenches.

8. LIFTING EQUIPMENT

M30 4ton collard eye bolts with 2 x half nuts and 1 plain washer per eye.
 M24 2.5ton collard eye bolts with 2 x half nuts and 1 plain washer per eye.
 1 x 4 leg lifting chain that's capable of lifting a minimum of the unit's weight on just two legs.
 These legs need to be long enough to reach the 4 corners of the unit when lying down.

9. EYE BOLTS

We recommend drop forged-table 1-BS4278 eyes, these should always be fitted using 2 half nuts, which are locked against each other and not the steel work so as to let the eye bolts rotate as the unit is turned.



M30 4ton eye, Bearward part number
6400013100

M30 flt washer, Bearward part number
3300007000

M30 half nut, Bearward part number
3300006900



M24 2.5ton eye, Bearward part number
6400001200

M24 flt washer, Bearward part number
3300000900

M24 half nut, Bearward part number
3300007100

10. LIFTING POINTS

10.1. Many of our units all use the same type of and combination of lifting eyes and brackets, most of which will be labelled as follows.



10.2. The above label tells you that this lifting point is to be used for a straight lift and for turning the radiator. These points should be fitted with an M30 4ton collard eye bolt.



10.3. When a lifting hole is fitted with this label it should only be used for a straight lift and should be fitted with M24 2.5ton collard eye bolts.